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Abstract

Being deaf or hard of hearing can result in exclusion due to a lack of understanding in communication; leading to feelings of alienation. Sign language allows for those of hard of hearing to learn, socialize and work like anyone else, bridging gaps on an individual level and fostering inclusivity. Our study integrates immersive Virtual Reality (VR) environments combined with an Artificial Intelligence (AI) to enhance the learning experience and retention of American Sign Language (ASL). Additionally, we are testing how distance affects engagement by creating far and close scenes for each condition. In a pilot study (N=8), participants in and outside the lab were asked to sign phrases in either a recreated conference room to an AI avatar or in a room with a TV and instructions. Results favored the close and far conference room with the AI avatar, showing higher speed and accuracy for test scores in the close scene and higher engagement times for the far scene. Our aim is to provide an engaging and responsive way to learn ASL by creating a safe and enjoyable environment that encourages learning and retention through leveraging the potential of VR and AI.

Background

- Importance of learning ASL
 - Effective communication with the Deaf Community
 - Cultural awareness and inclusion.
 - Valuable skill that can open various career opportunities.
 - Enhanced special reasoning and ability to interpret body language
- Why VR and AI?
 - Immersive and interactive learning.
 - Safe and controlled environment.
 - Diverse and customizable scenarios.
 - Immediate feedback for faster learning.

Method

Development Tools: Unity, C#, R, RStudio, OpenAI GPT API, Valid Avatar Library, Meta Quest 1 Headset, MiVRy

Figure 1a. HCI conference room

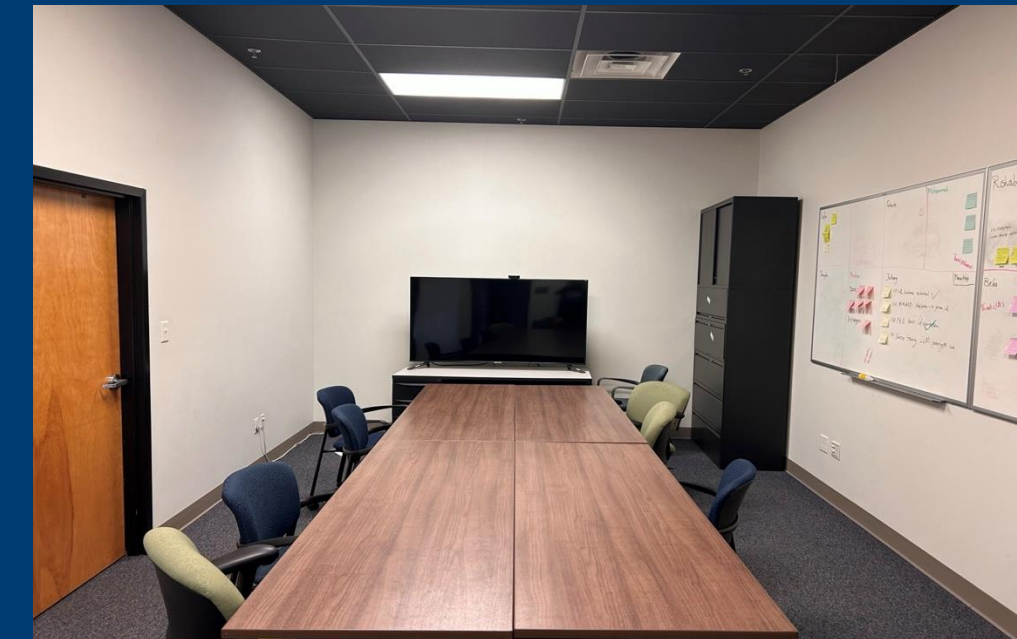


Figure 1c. Participant's perspective in recreated conference room (far condition)

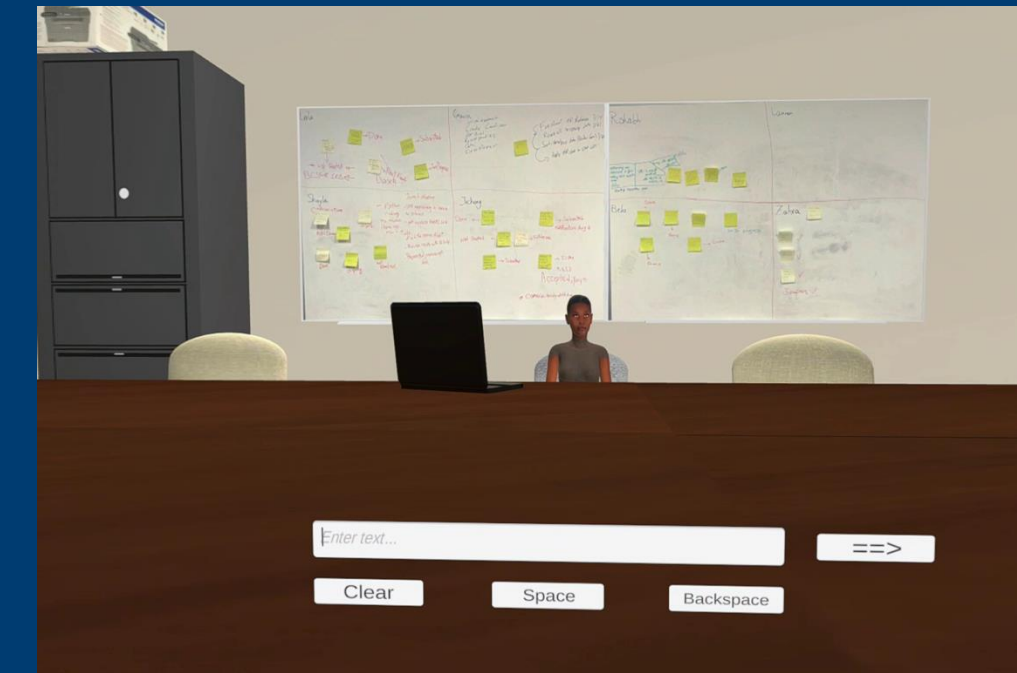


Figure 1b. Virtual reality conference room

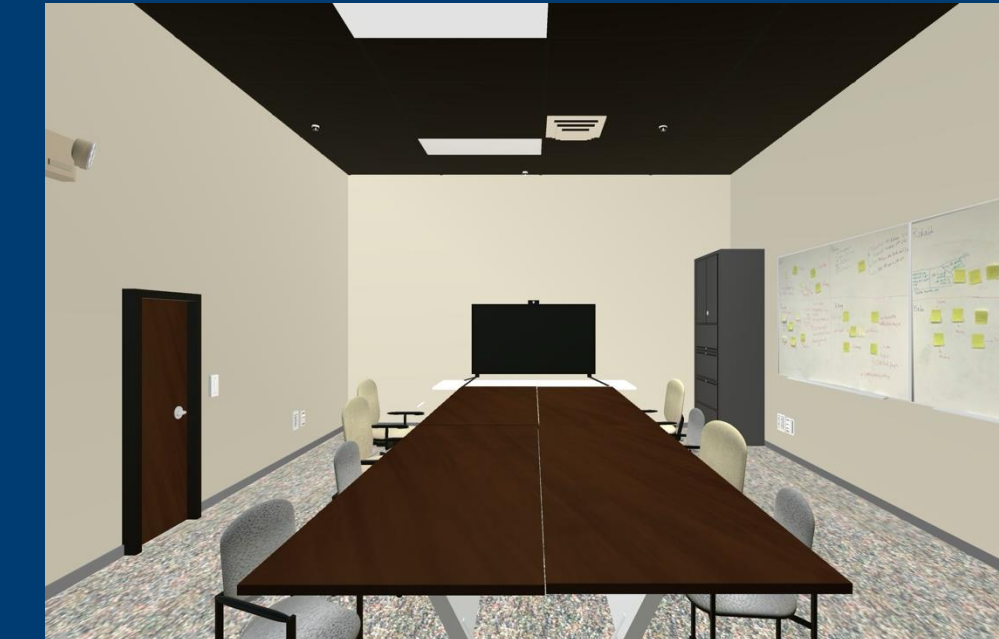


Figure 1d. Participant's perspective in default room (close condition)



Figure 1e. Participant gesturing to AI avatar in recreated conference room (close condition)

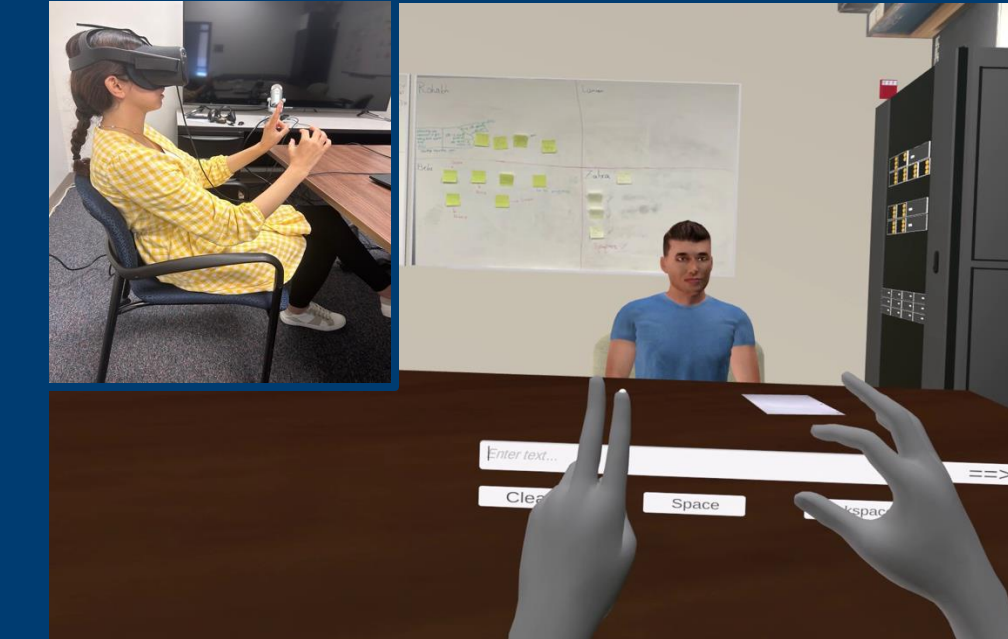


Figure 1g. Participant gesturing to AI avatar in recreated conference room (far condition)

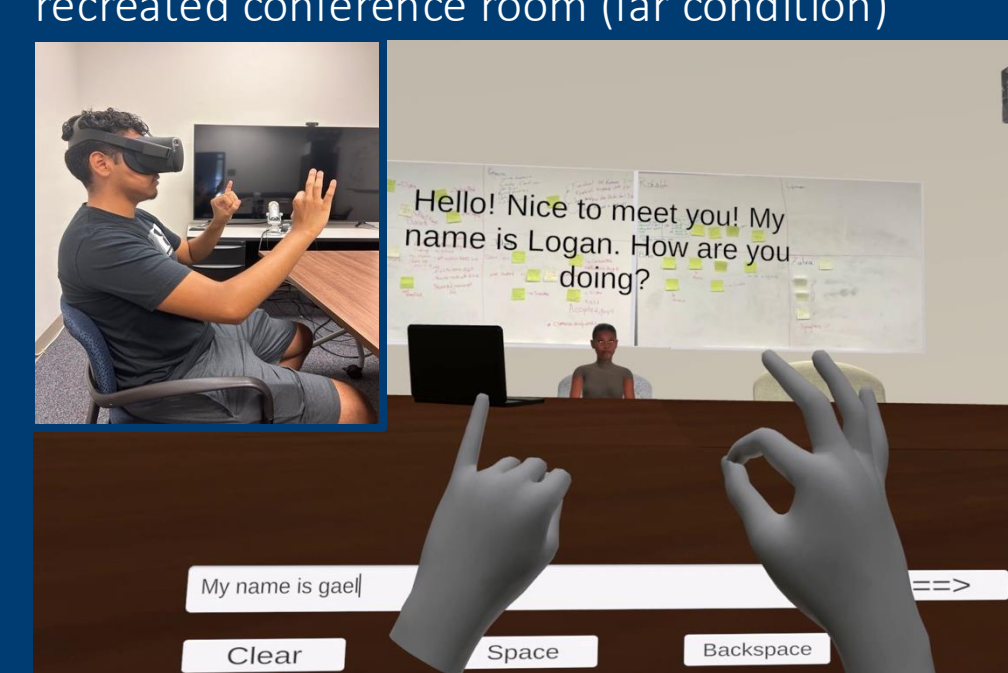


Figure 1f. Participant gesturing to a TV with instructions in a default room (close condition)

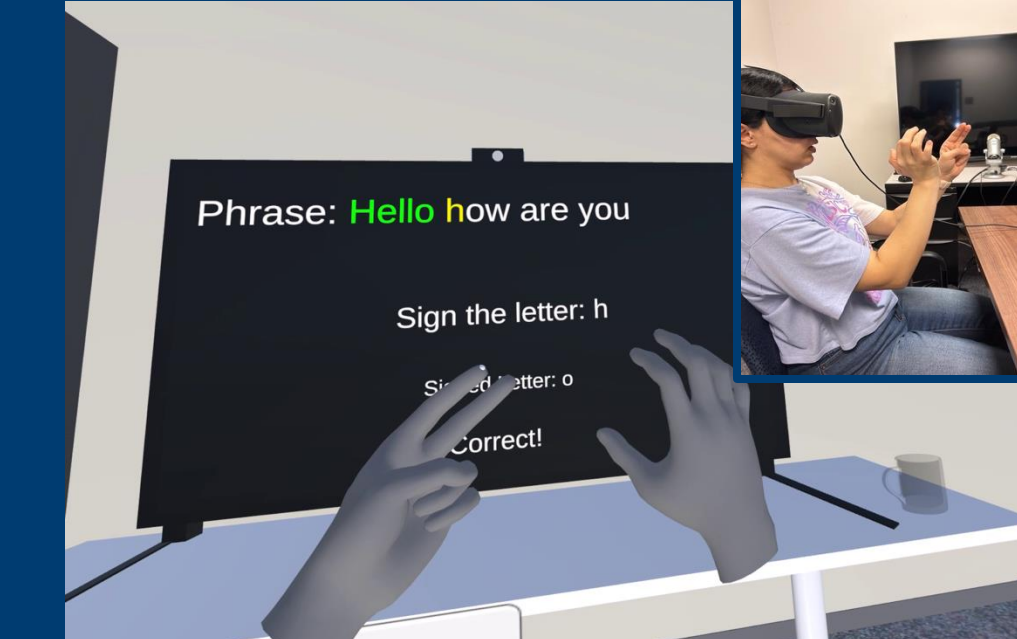
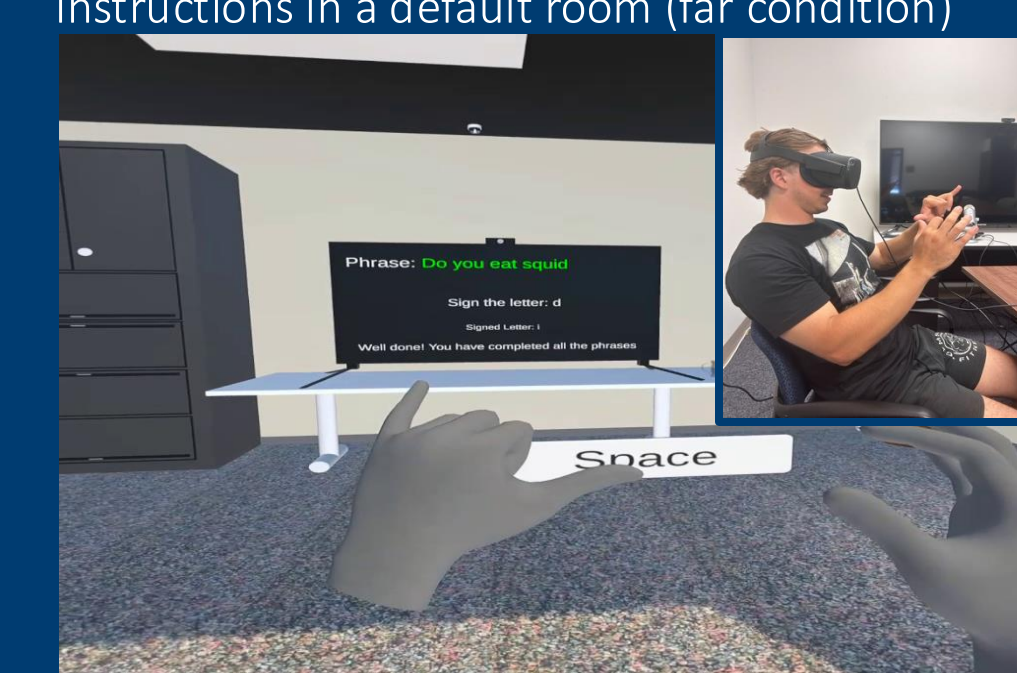


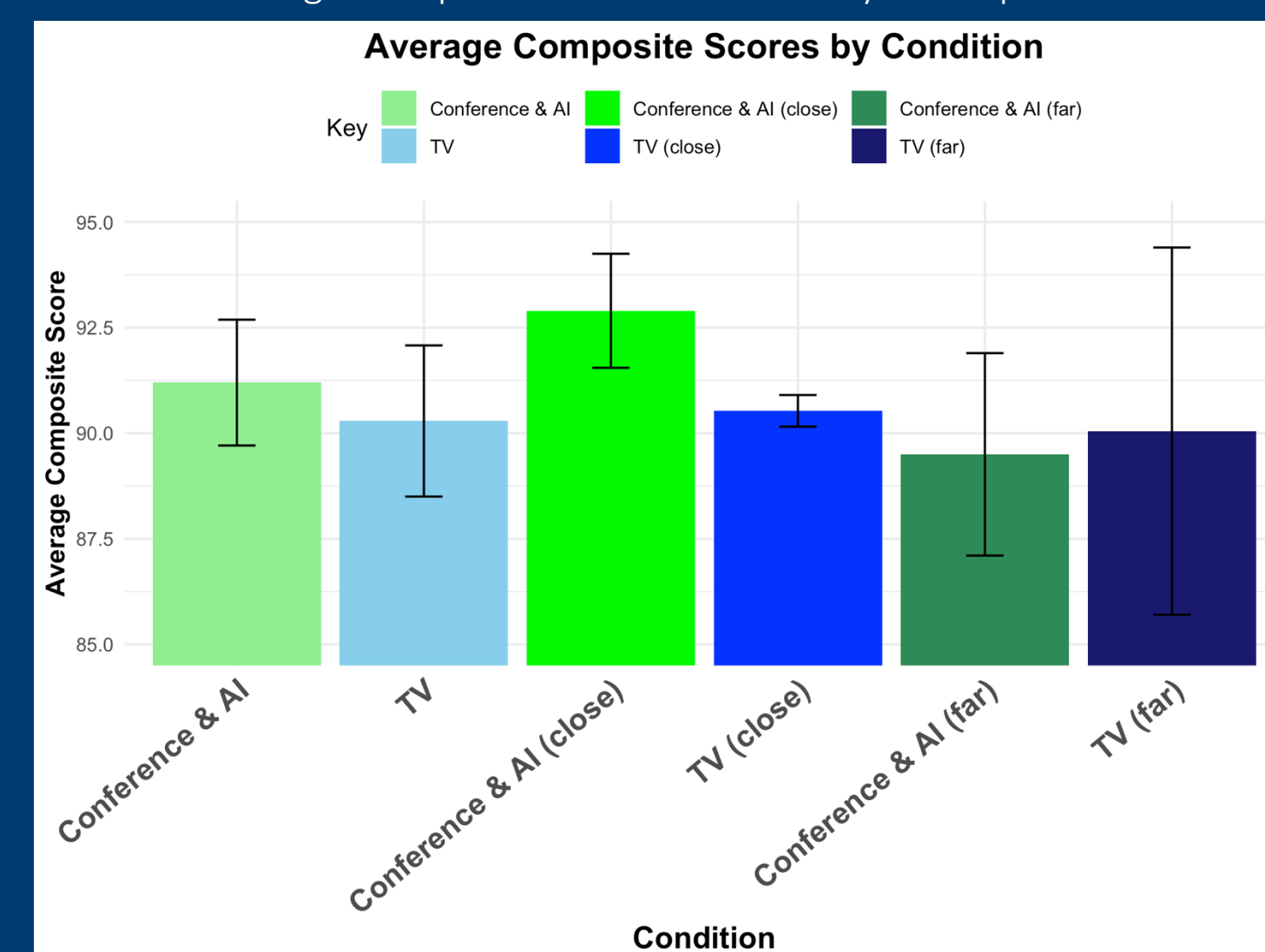
Figure 1h. Participant gesturing to a TV with instructions in a default room (far condition)



Pilot Test:
8 participants (4M, 4F) with no prior knowledge of the ASL alphabet practiced the alphabet in their respective conditions and were then tested for speed and accuracy. Participants completed a survey after the test using 4-point Likert scales regarding user experience. Participants would take the same test again the next day.

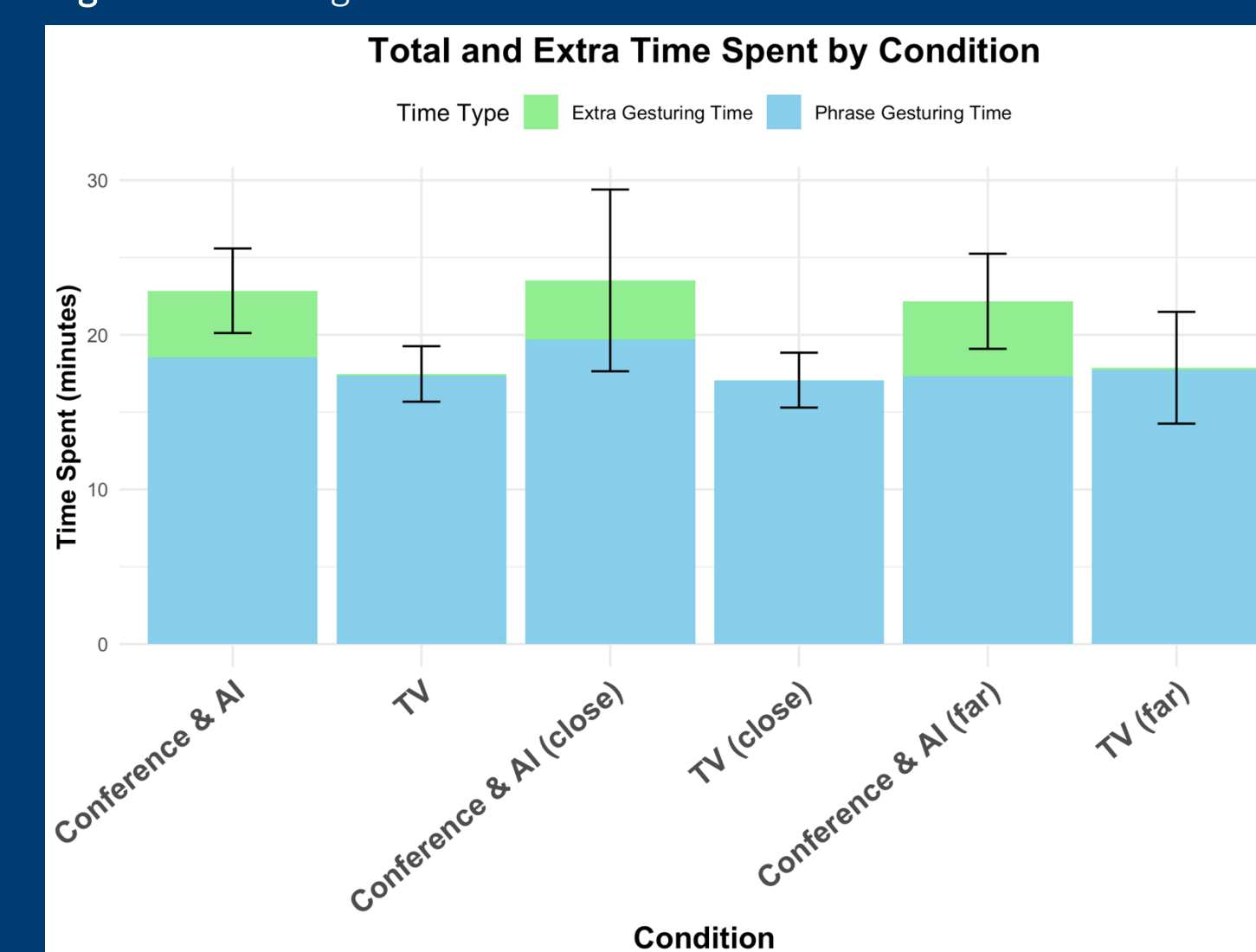
Results

Figure 2a. Average composite score of all conditions. Score was calculated using the equation: Score = Accuracy - α x Speed.



- The **Conference & AI** group achieved a higher average composite score compared to the TV group.
- In both **Conference & AI** and **TV** groups, participants in the close condition had higher scores than the far condition.

Figure 2b. Average total time of each condition.



- Both close and far **Conference and AI** conditions engaged in extra gesturing after finishing the assigned phrases.
- Most participants continued the conversation with the AI avatar.

Figure 2c. Survey Data: 1 - Strongly Disagree, 4 - Strongly Agree



- The **Conference & AI** conditions provided a more immersive and engaging environment compared to the TV conditions.
- Some participants reported low frustration levels while others experienced higher levels.
- Both groups equally enjoyed the VR experience and recommended their respective condition as a way to practice the ASL alphabet.

Conclusions

- Immersive environments with AI elements allowed for users to go beyond required practice in an engaging manner that positively affects learning.
- Interactive elements benefit from strategic placement. Closer materials enhance learning and retention, while farther materials boost engagement and interest.
- Gesture detection needs to be more accurate to curb frustration and disinterest.
- Conducting a larger study with a bigger participant base will be essential for generalizing the findings of this study.

Future Work

- Right hand gesture control
- Different environments and context for different scenarios
- A larger sample size study with diverse participants
- Full word gestures/two hand gestures
- Make the gesture detection more accurate.

Acknowledgments

This poster was supported by the Delaware INBRE program, with a grant from the National Institute of General Medical Sciences – NIGMS (P20GM103446) from the National Institutes of Health.

I'd like to thank Gavin Caulfield for the external help and support in the creation and foundation of the project

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